

Electric Circuits

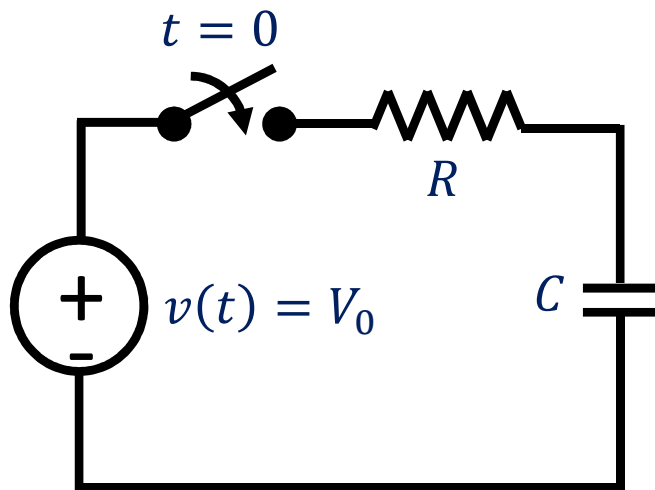
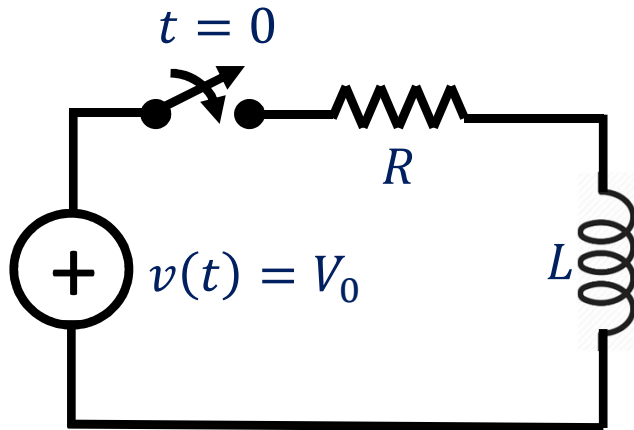


*Series RL and RC Circuits
Tutorial Problems*

Tentative Lecture Plan

L	LDay-1	LDay-2	Topics	T	TDay
1	FRI, 04 AUG	MON, 07 AUG	Basic Definitions, Voltage & Current Sources, KCL & KVL	1	THU, 10 AUG
2	TUE, 08 AUG	FRI, 11 AUG	Nodal Analysis: Independent and Dependent Current Sources	2	SAT, 19 AUG
3	MON, 14 AUG	THU, 17 AUG	Nodal Analysis: Current and Voltage Sources		
4	FRI, 18 AUG	MON, 21 AUG	Mesh Analysis: Voltage and Current Sources	3	THU, 24 AUG
5	TUE, 22 AUG	FRI, 25 AUG	Network Theorems: Superposition Theorem	R	SAT, 26 AUG
SUN, 27 AUG : Quiz-1 : Lectures 1-5					
6	MON, 28 AUG	TUE, 29 AUG	Network Theorems: Thevenin's and Norton's Theorem	4	THU, 31 AUG
7	MON, 04 SEP	TUE, 05 SEP	Network Theorems: Maximum Power Transfer Theorem	5	WED, 06 SEP
8	FRI, 08 SEP	SAT (Mo), 09 SEP	Series RL and RC Circuits		
9	MON, 11 SEP	TUE, 12 SEP	Series RL and RC Circuits: Tutorial Problems	6	THU, 14 SEP
10	FRI, 15 SEP	SAT (Tu), 16 SEP	Series RL, RC and RLC Circuits Response to Constant, Sinusoid and Exponential Sources		
WED, 20 SEP: MidSem : Lectures 1 – 10					

Response of First Order Linear Circuits



$$i_L(t) = \frac{V_0}{R} \{1 - e^{-\frac{R}{L}t}\} + i_L(0^+)e^{-\frac{R}{L}t}$$

$$x(t) = x(\infty) \{1 - e^{-\frac{t}{\tau}}\} + x(0^+)e^{-\frac{t}{\tau}}$$

$$v_c(t) = \underbrace{V_0 \{1 - e^{-\frac{t}{RC}}\}}_{\text{Forced Response}} + \underbrace{v_c(0^+)e^{-\frac{t}{RC}}}_{\text{Natural Response}}$$

Forced
Response

Natural
Response

Solving First Order Linear Circuits

- Consider evaluating the branch variable $x(t)$ of a first order linear circuit with a single capacitor C or a single inductor L . At least one branch gets modified while a switch opens or closes at $t = 0$. The circuit has constant voltage and/or current sources.
- The solution of the branch variable $x(t)$ can be obtained as $x(t) = x_f(t) + x_n(t)$
- Draw the circuit at $t = 0^-$ with capacitor replaced by an open circuit or inductor replaced by a short circuit. Solve for $v_c(0^-)$ or $i_L(0^-)$ prior to switch action.
- Draw the circuit post switch action to solve for $x(0^+)$. Replace capacitor with a voltage source $v_c(0^-) = v_c(0^+)$ or an inductor with a current source $i_L(0^-) = i_L(0^+)$.

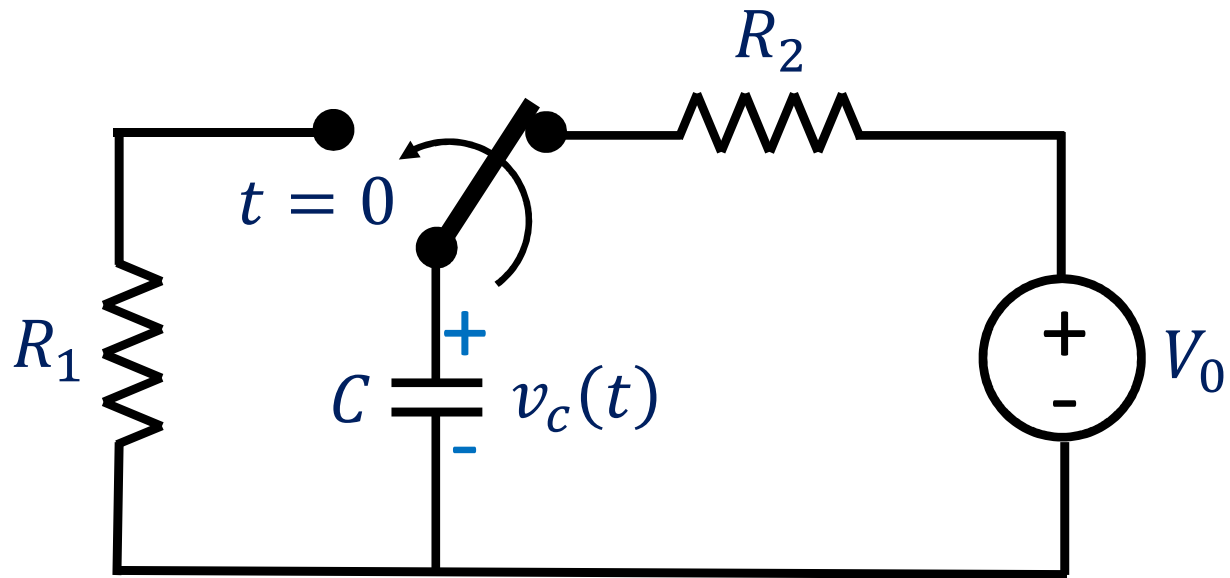
Solving First Order Linear Circuits

- Draw the steady state equivalent of the circuit by replacing the capacitor with an open circuit or the inductor with a short circuit. Solve for $x(\infty)$.
- Reduce the entire circuit to a series RL or RC circuit by evaluating the Thevenin's equivalent circuit (with source V_{OC} and resistance R_{TH}). Here, the capacitor or inductor is connected across the load terminals. Thus, the time constant can be obtained as either $\tau_{RC} = R_{TH}C$ (for RC series) or $\tau_{RL} = \frac{L}{R_{TH}}$.
- The solution of the branch variable $x(t)$ is obtained as

$$x(t) = x_f(t) + x_n(t)$$

$$x(t) = x(\infty)\left(1 - e^{-\frac{t}{\tau}}\right) + x(0^+) e^{-\frac{t}{\tau}}$$

RC Circuit



$$R_1 = 80\Omega$$

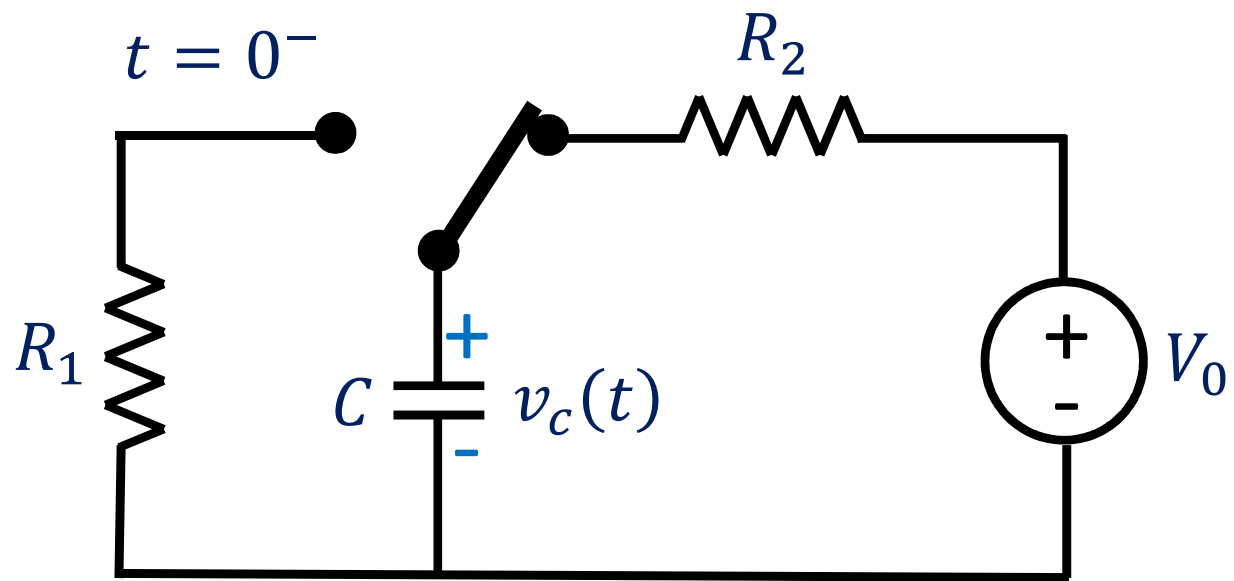
$$R_2 = 70\Omega$$

$$C = 2\mu F$$

$$V_0 = 50V$$

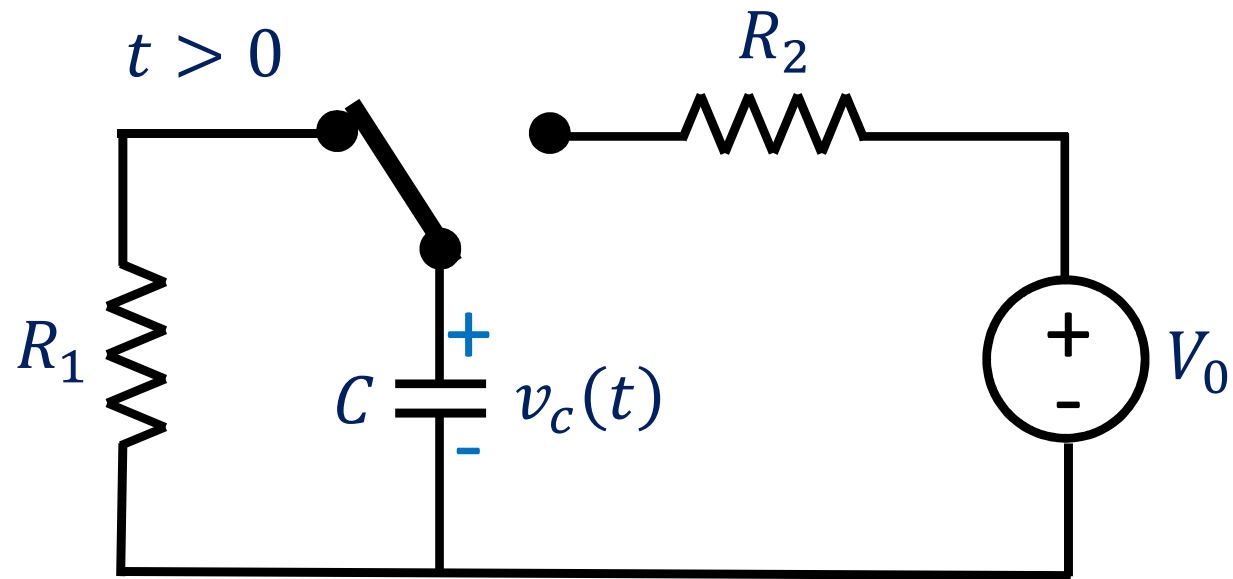
$$v_c(t = 160\mu s) = ?$$

RC Circuit



$$v_c(0^-) = V_0 = 50V$$

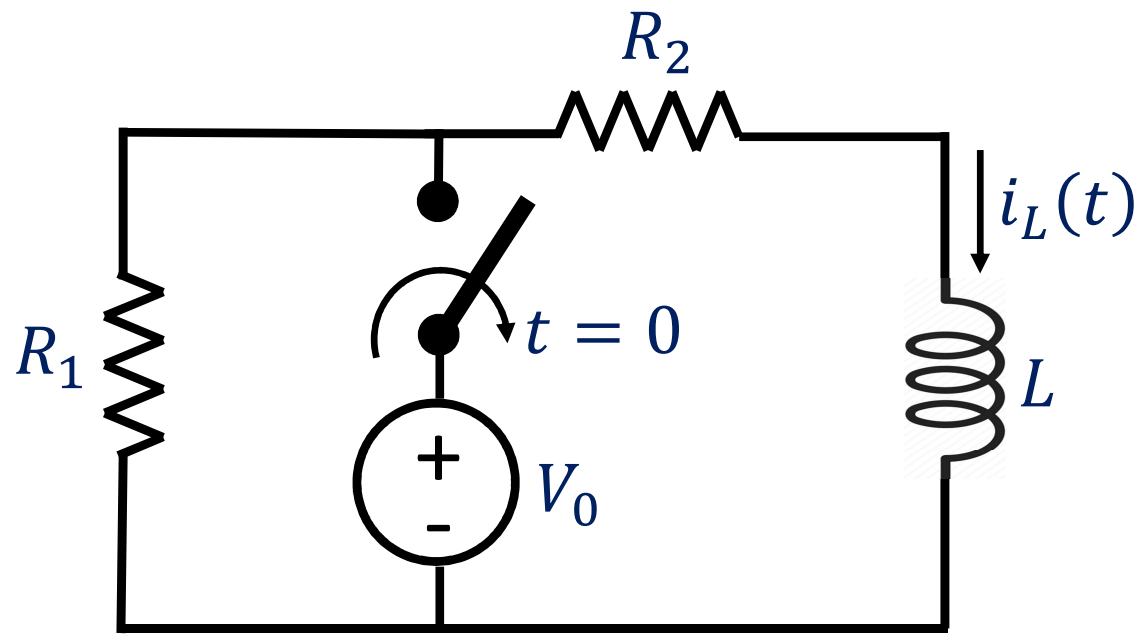
RC Circuit



$$\tau = R_1 C = 80 \times 2 \times 10^{-6} \text{ s} = 160 \mu\text{s}$$

$$v_c(t) = v_c(0^+) e^{-\frac{t}{\tau}} \quad \Rightarrow \quad v_c(160 \mu\text{s}) = 50 e^{-\frac{160}{160}} \text{ V} = 18.39 \text{ V}$$

RL Circuit



$$R_1 = 6\Omega$$

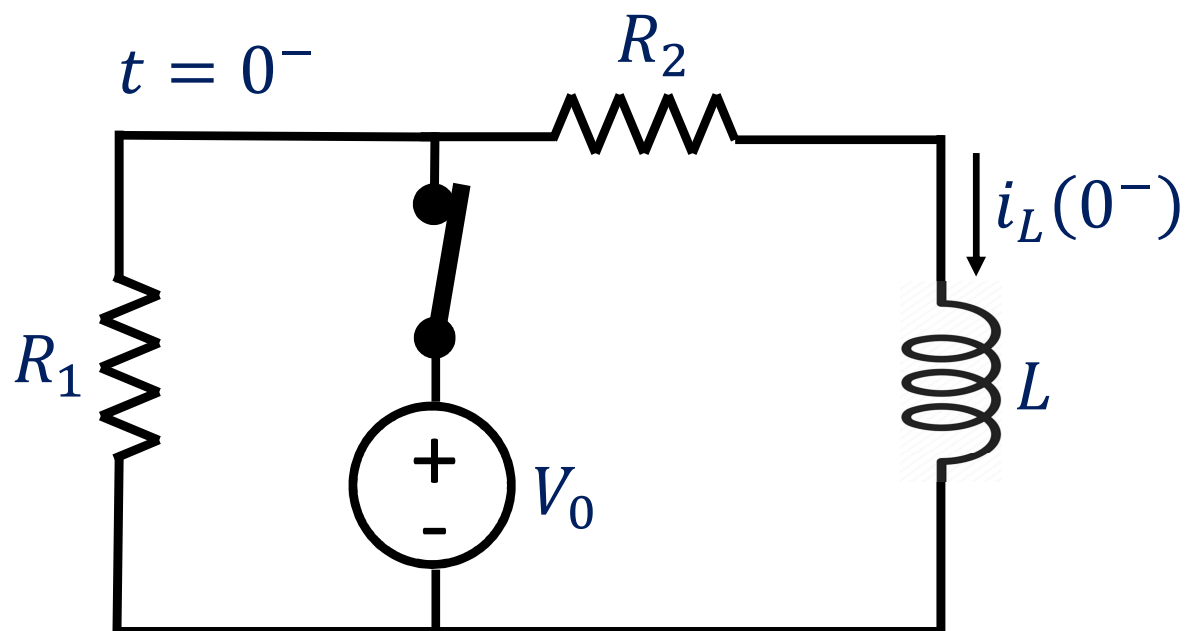
$$R_2 = 4\Omega$$

$$L = 5H$$

$$V_0 = 10V$$

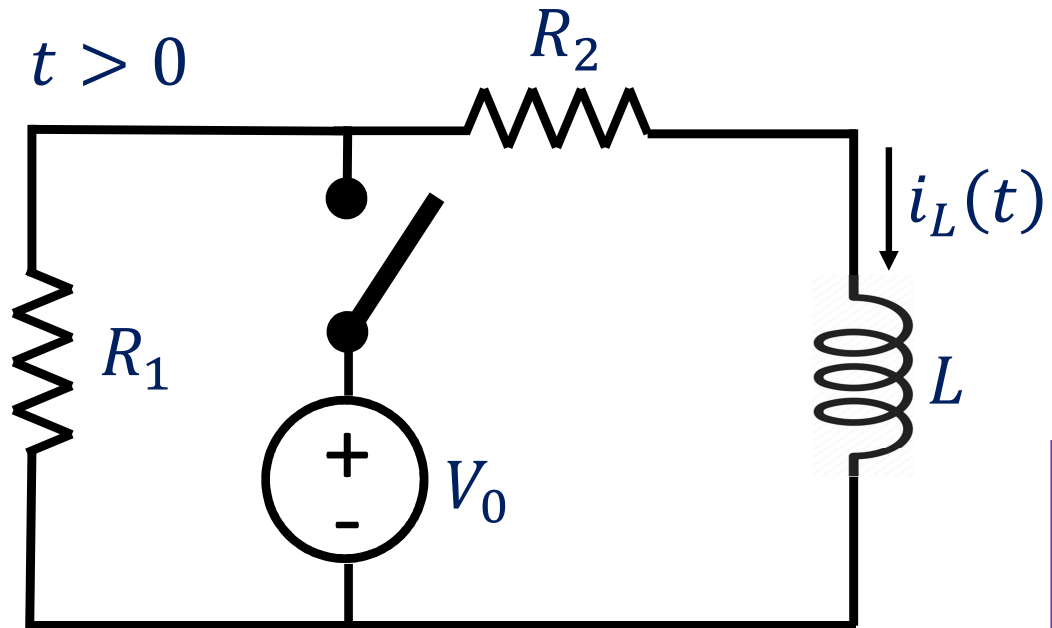
$$i_L(t) = ?$$

RL Circuit



$$i_L(0^-) = \frac{V_0}{R_2} = \frac{10}{4} \text{ A} = 2.5 \text{ A}$$

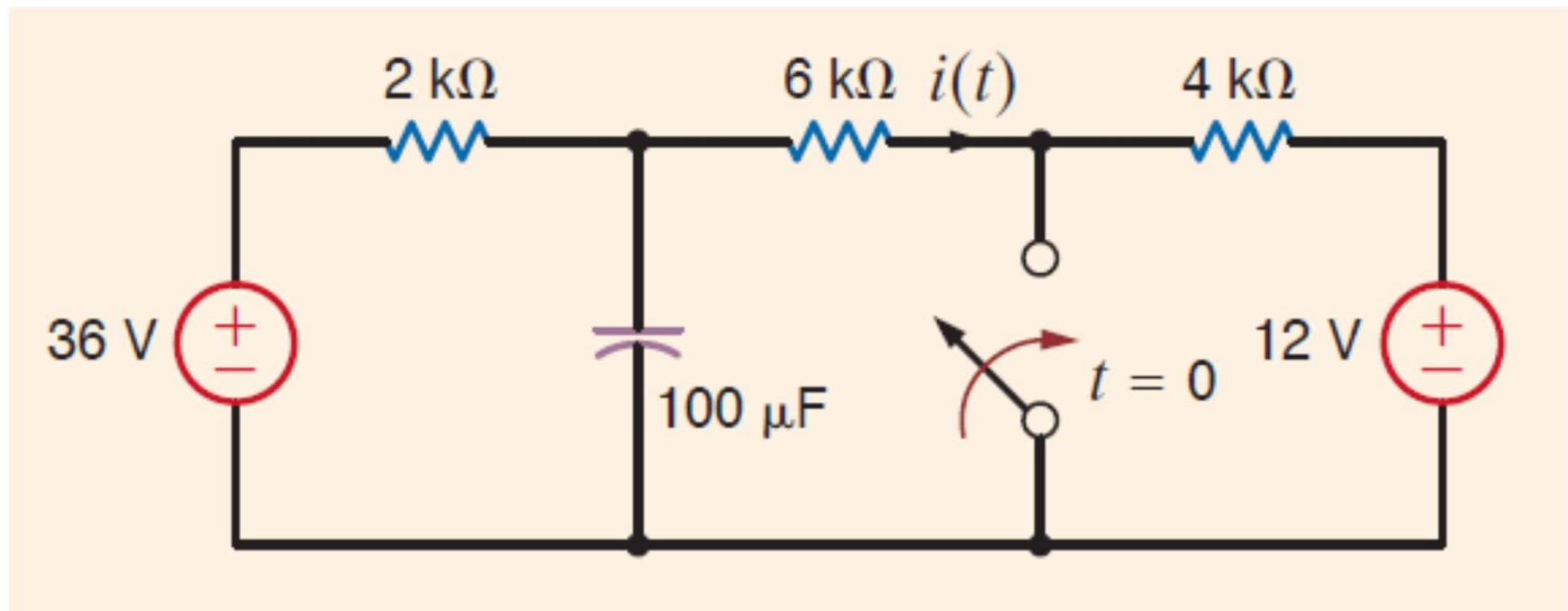
RL Circuit



$$\tau = \frac{L}{R_1 + R_2} = \frac{5}{6 + 4} = 0.5 \text{ s}$$

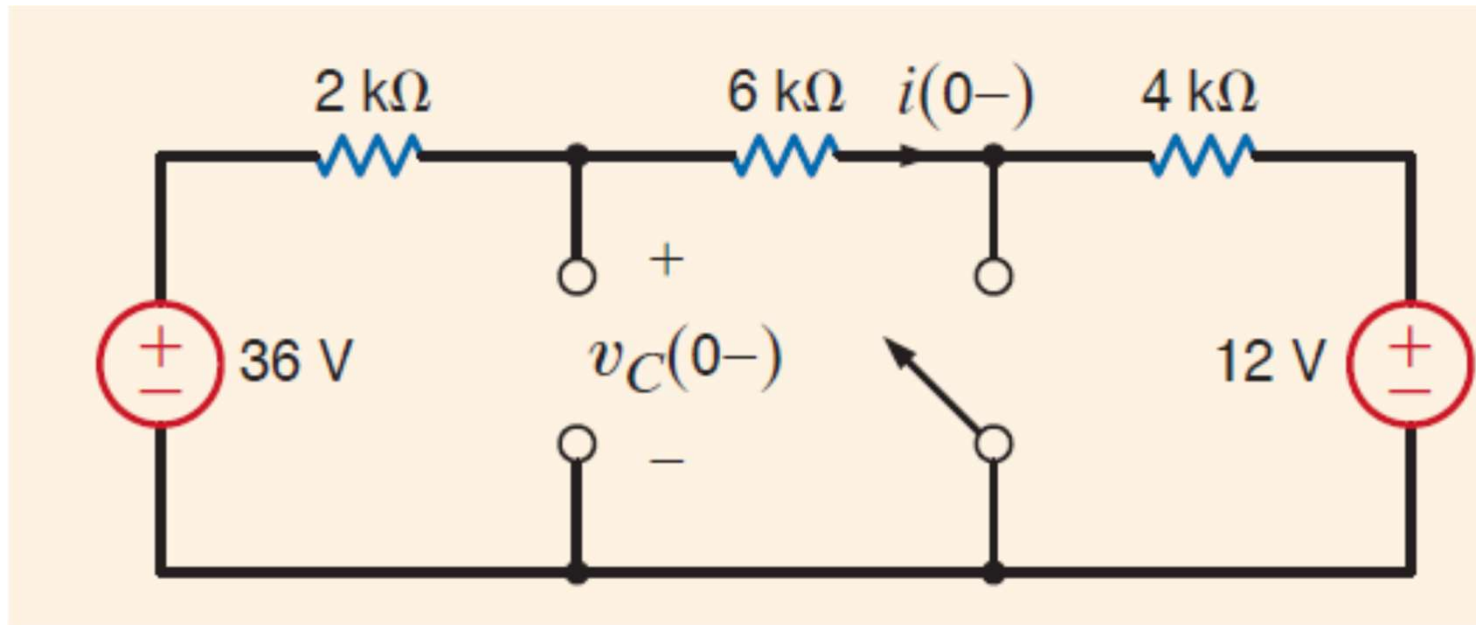
$$i_L(t) = i_L(0^+) e^{-\frac{t}{\tau}} = 2.5 e^{-2t} \text{ A}$$

RC Circuit



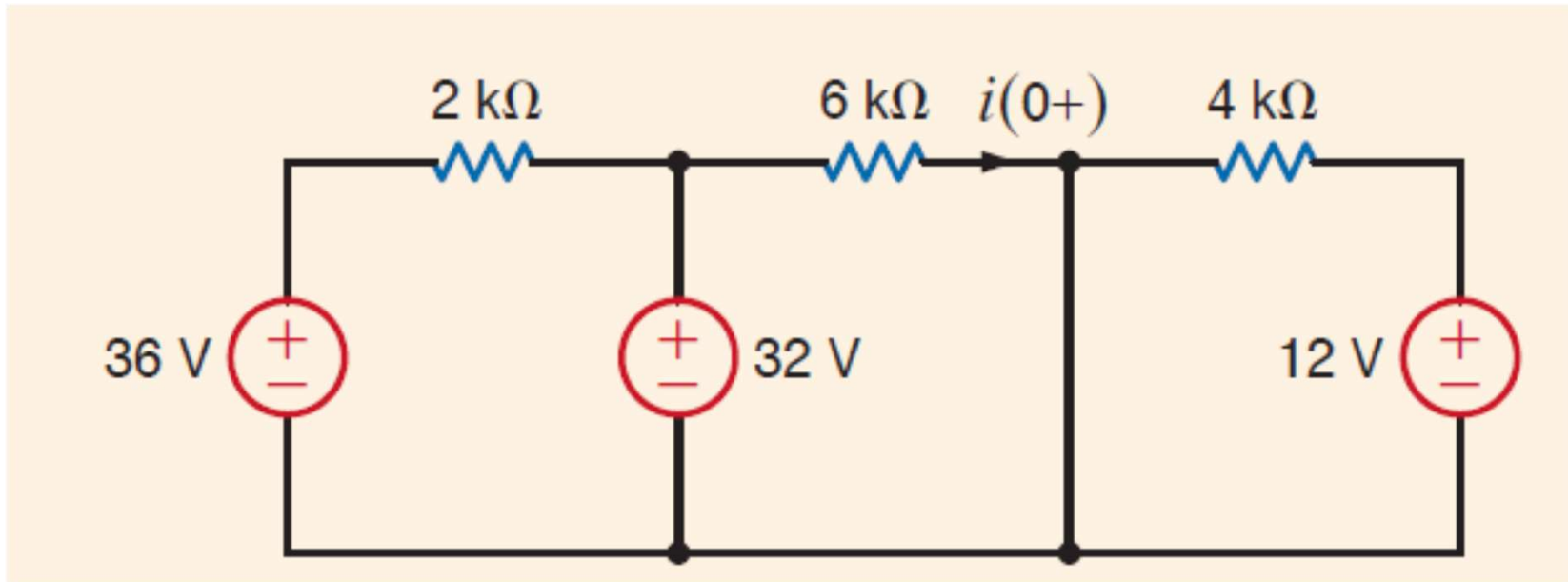
$$i(t) = ?$$

RC Circuit



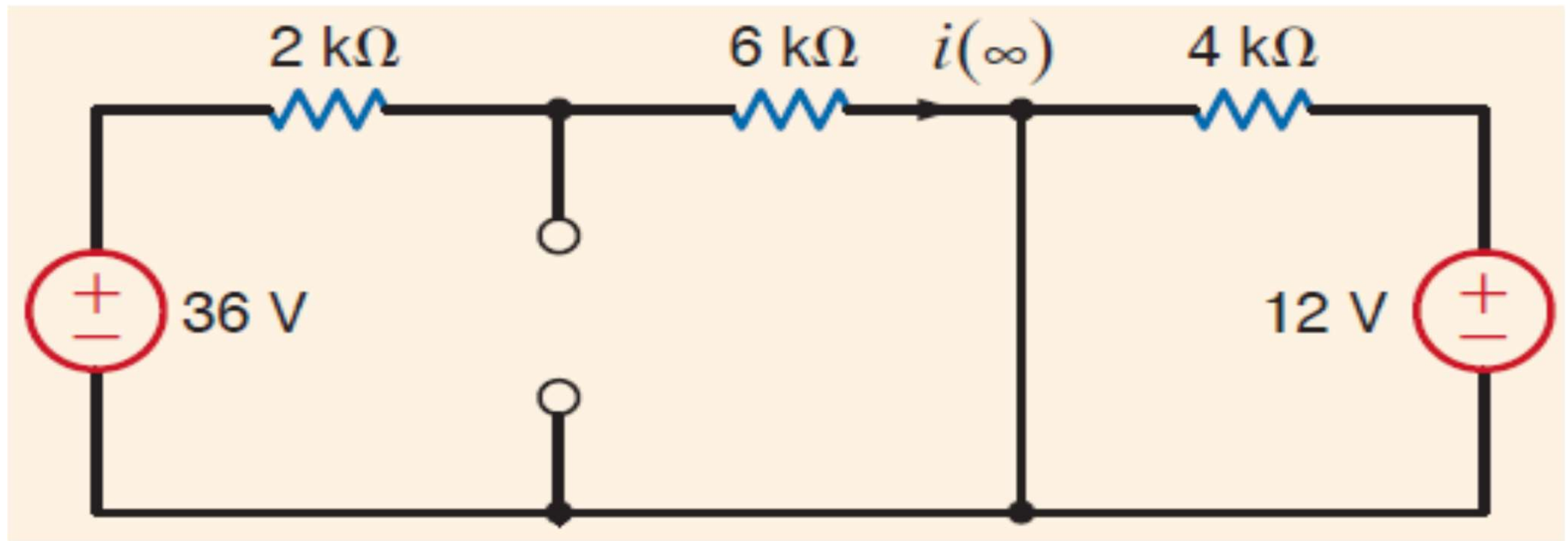
$$v_C(0^-) = v_C(0^+) = \frac{36 \times 10 + 12 \times 2}{12} = 32V$$

RC Circuit



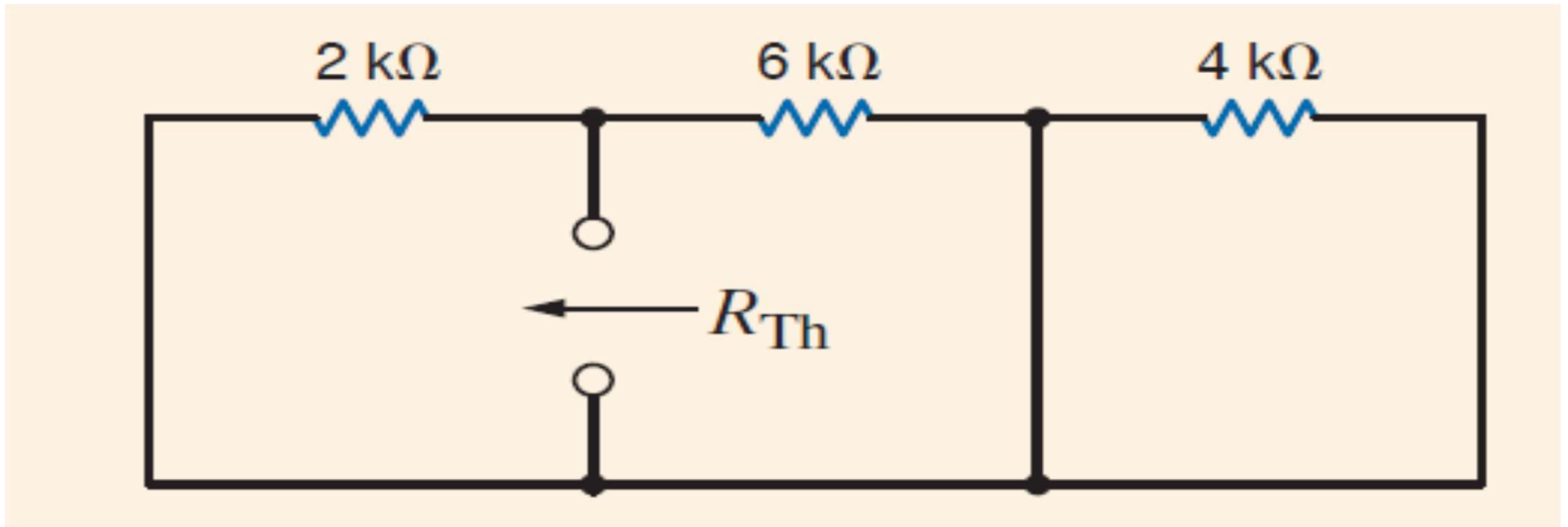
$$i(0^+) = \frac{32}{6 \times 10^3} = \frac{16}{3} \times 10^{-3} A$$

RC Circuit



$$i(\infty) = \frac{36}{(2 + 6) \times 10^3} = \frac{9}{2} \times 10^{-3} A$$

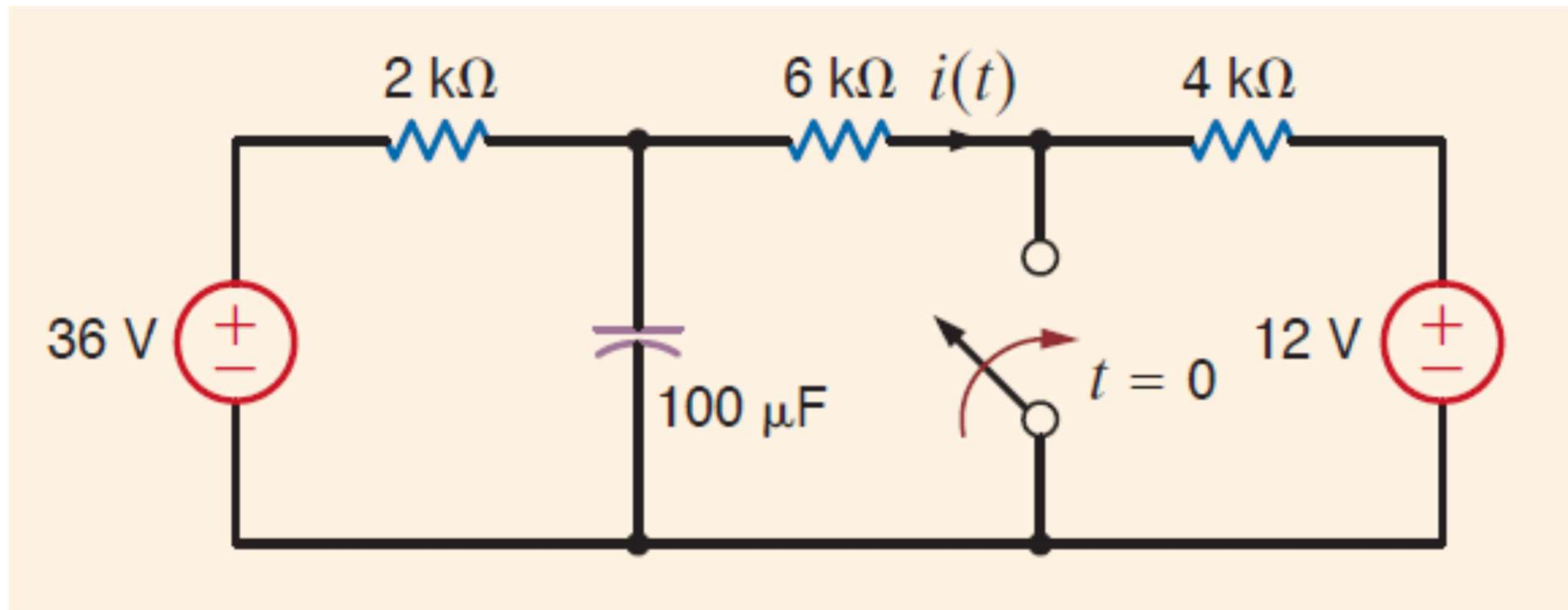
RC Circuit



$$R_{TH} = \frac{2 \times 6}{2 + 6} 10^3 = \frac{3}{2} \text{ k}\Omega$$

$$\tau_{RC} = R_{TH} C = 0.15 \text{ s}$$

RC Circuit



$$i(t) = i(\infty) \left\{ 1 - e^{-\frac{t}{\tau_{RC}}} \right\} + i(0^+) e^{-\frac{t}{\tau_{RC}}} = \{ 4.5 (1 - e^{-\frac{t}{0.15}}) + 5.33 e^{-\frac{t}{0.15}} \} \text{mA}$$

RL Circuit

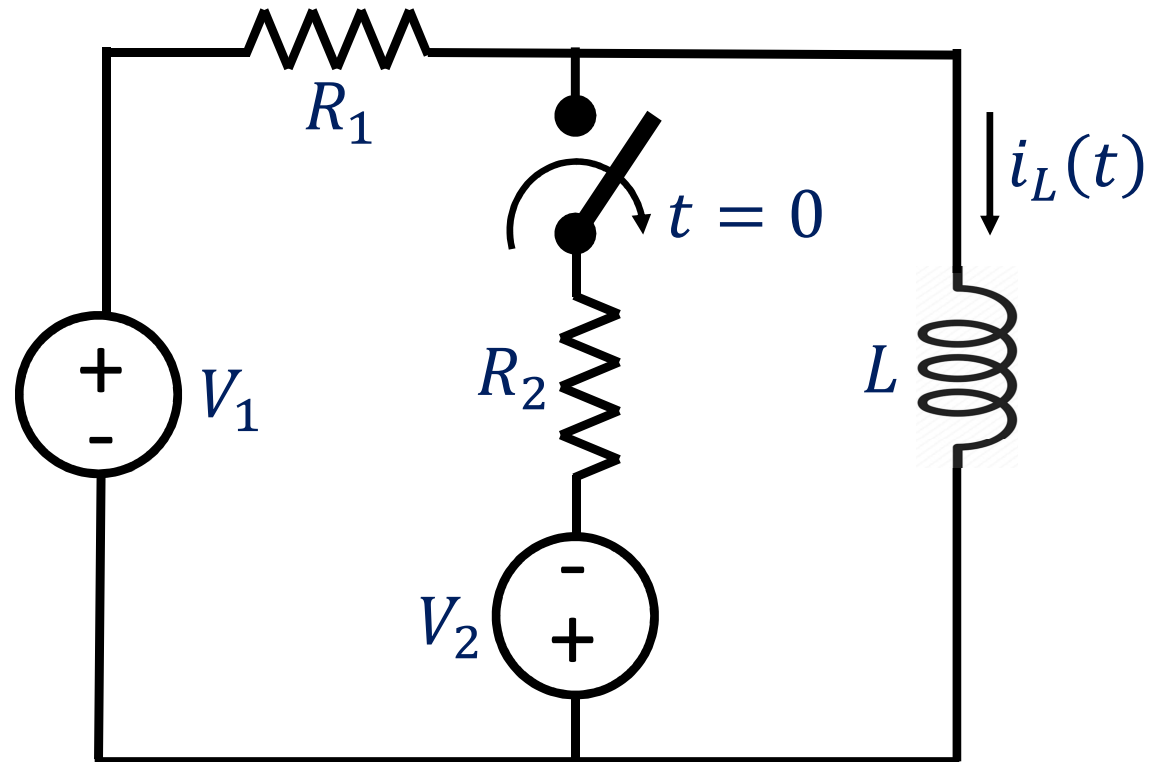
$$R_1 = 20\Omega$$

$$R_2 = 5\Omega$$

$$L = 0.5H$$

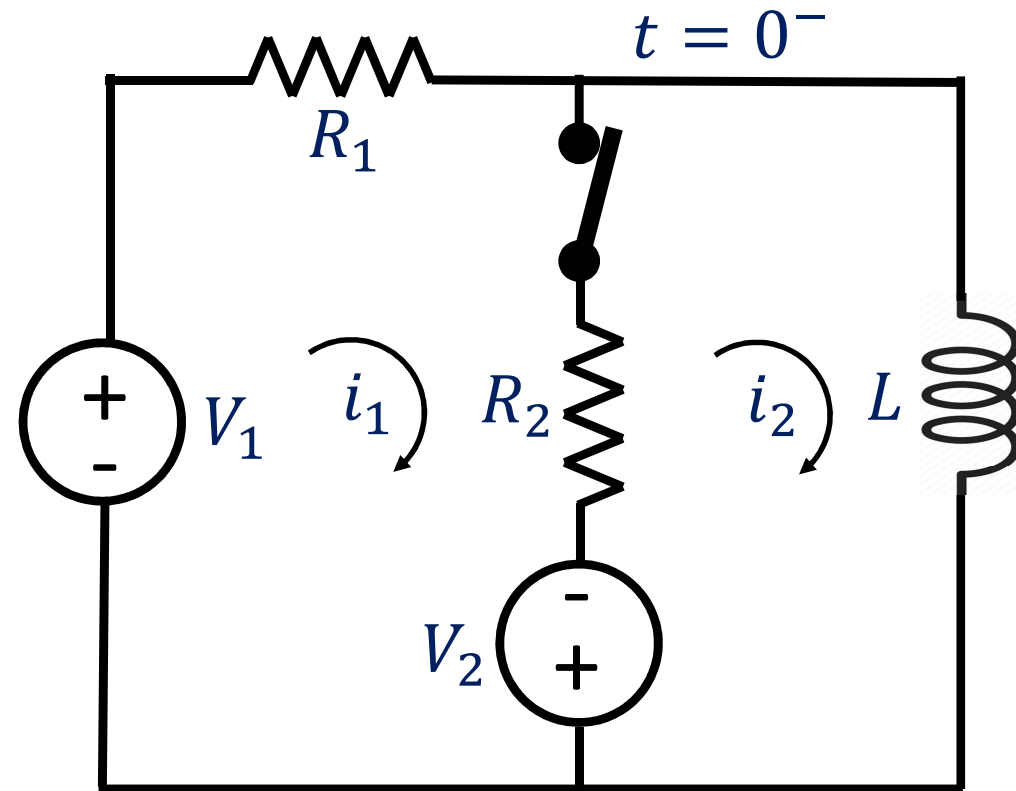
$$V_1 = 100V$$

$$V_2 = 100V$$



$$i_L(t) = ?$$

RL Circuit

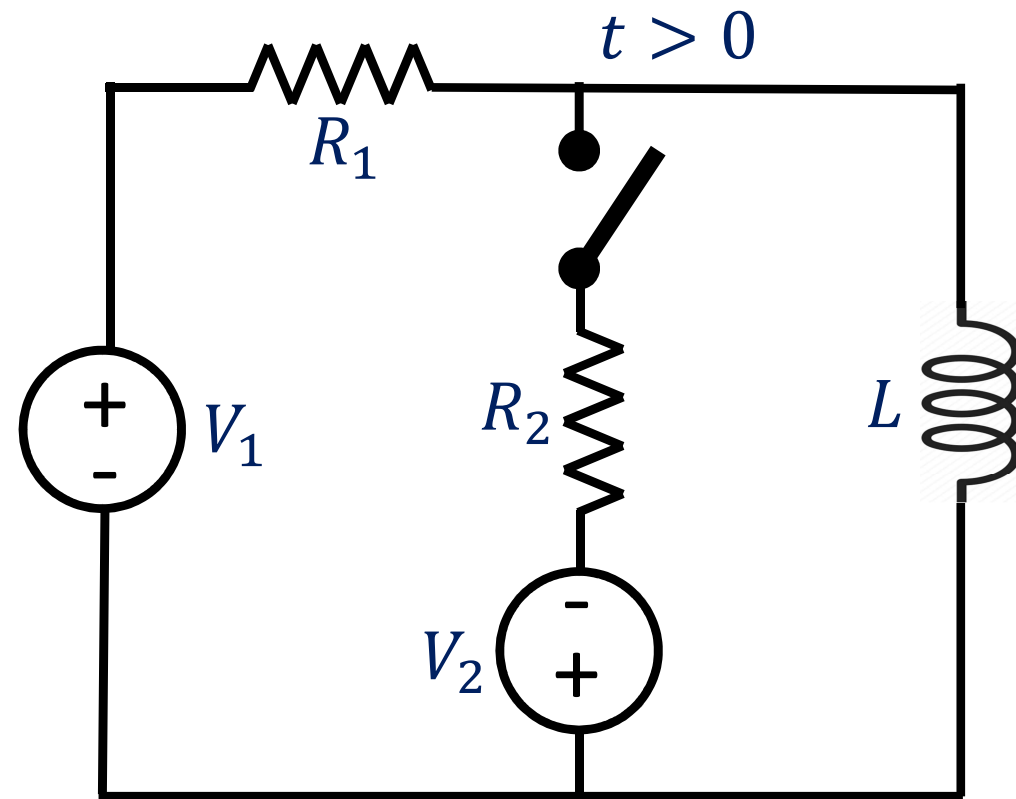


$$\begin{bmatrix} R_1 + R_2 & -R_2 \\ -R_2 & R_2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_1 + V_2 \\ -V_2 \end{bmatrix}$$

Solving Mesh Current Equations

$$i_2 = i_L(0^-) = -15 \text{ A}$$

RL Circuit

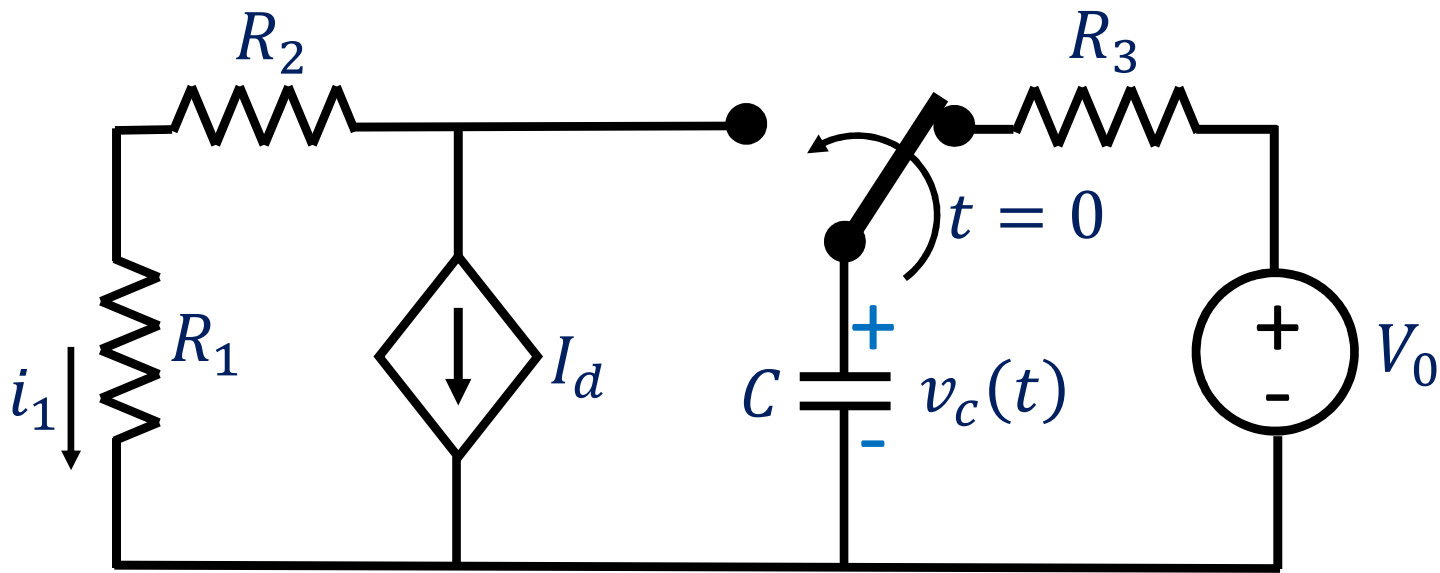


$$i_L(\infty) = \frac{V_1}{R_1} = \frac{100}{20} \text{ A} = 5 \text{ A}$$

$$\tau = \frac{L}{R_1} = \frac{0.5}{20} \text{ s} = \frac{1}{40} \text{ s}$$

$$i_L(t) = 5(1 - e^{-40t}) - 15e^{-40t}$$

RC Circuit with Dependent Source



$$R_1 = 20\Omega$$

$$R_2 = 10\Omega$$

$$R_3 = 30\Omega$$

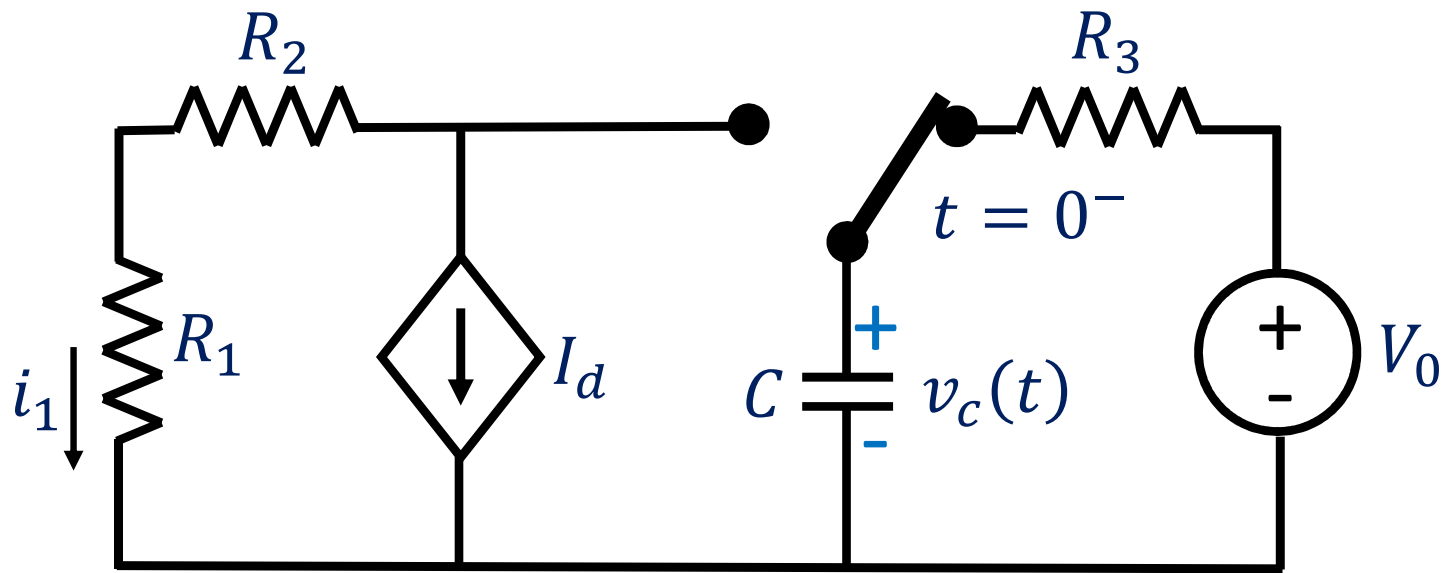
$$C = 1\mu F$$

$$V_0 = 2V$$

$$I_d = 1.5i_1$$

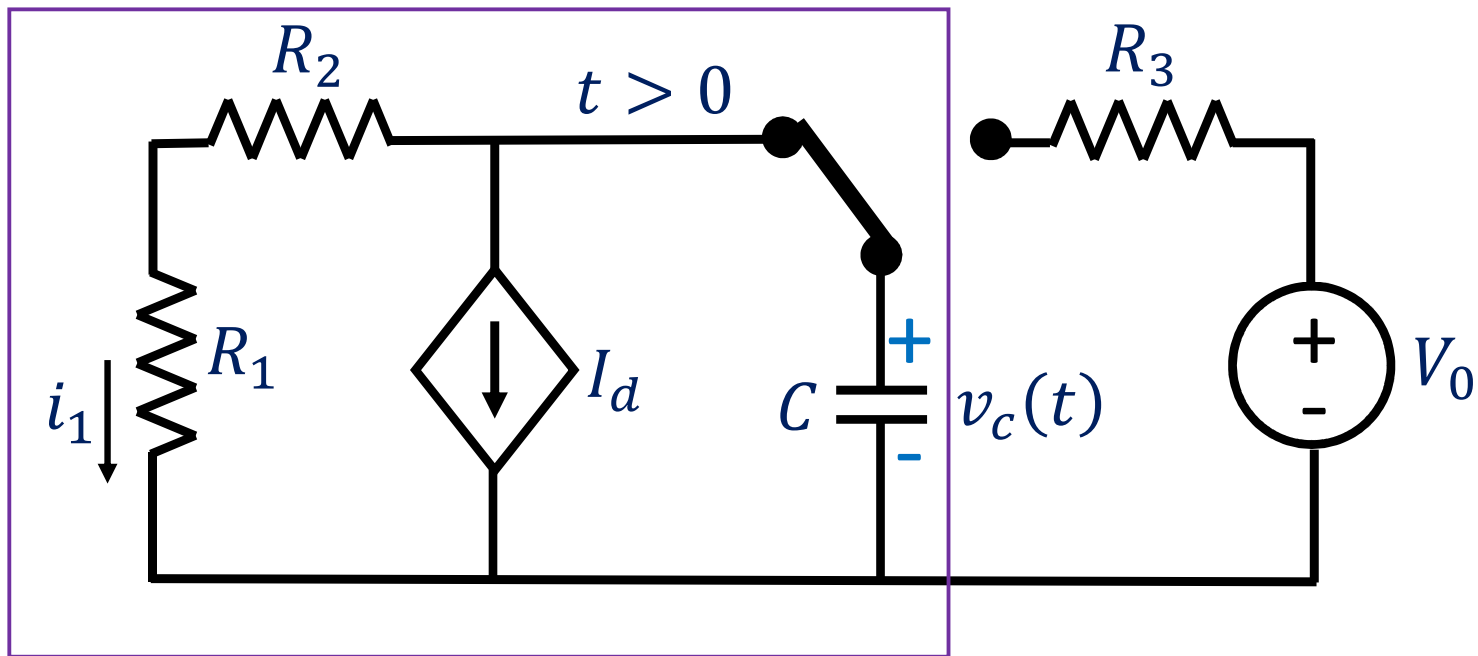
$$v_c(t) = ?$$

RC Circuit with Dependent Source



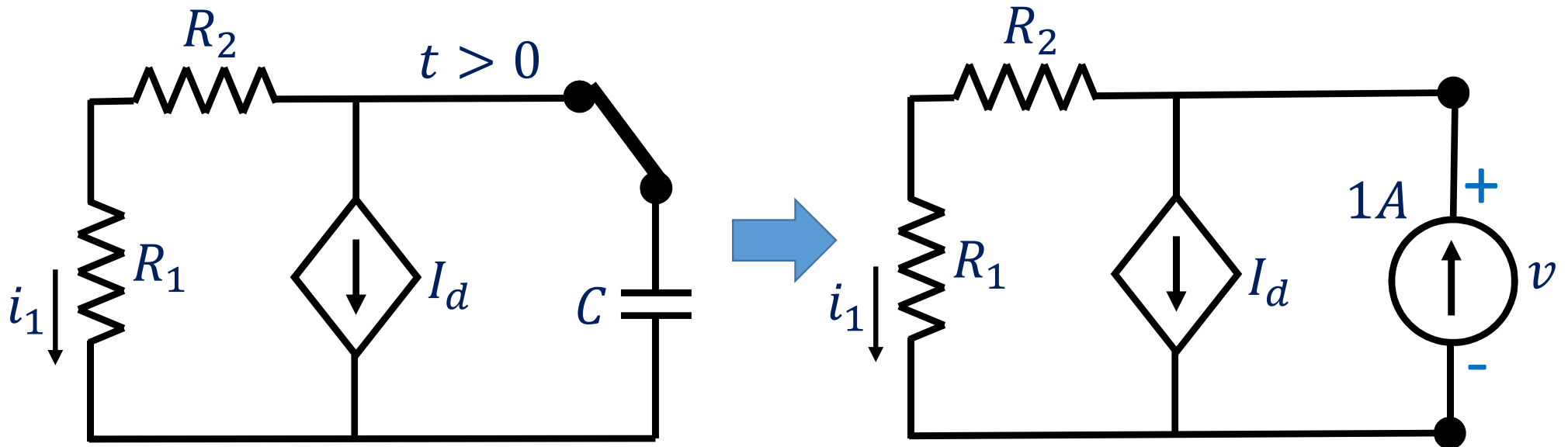
$$v_c(0^-) = V_0 = 2V$$

RC Circuit with Dependent Source



$$v_c(0^+) = v_c(0^-) = 2V$$

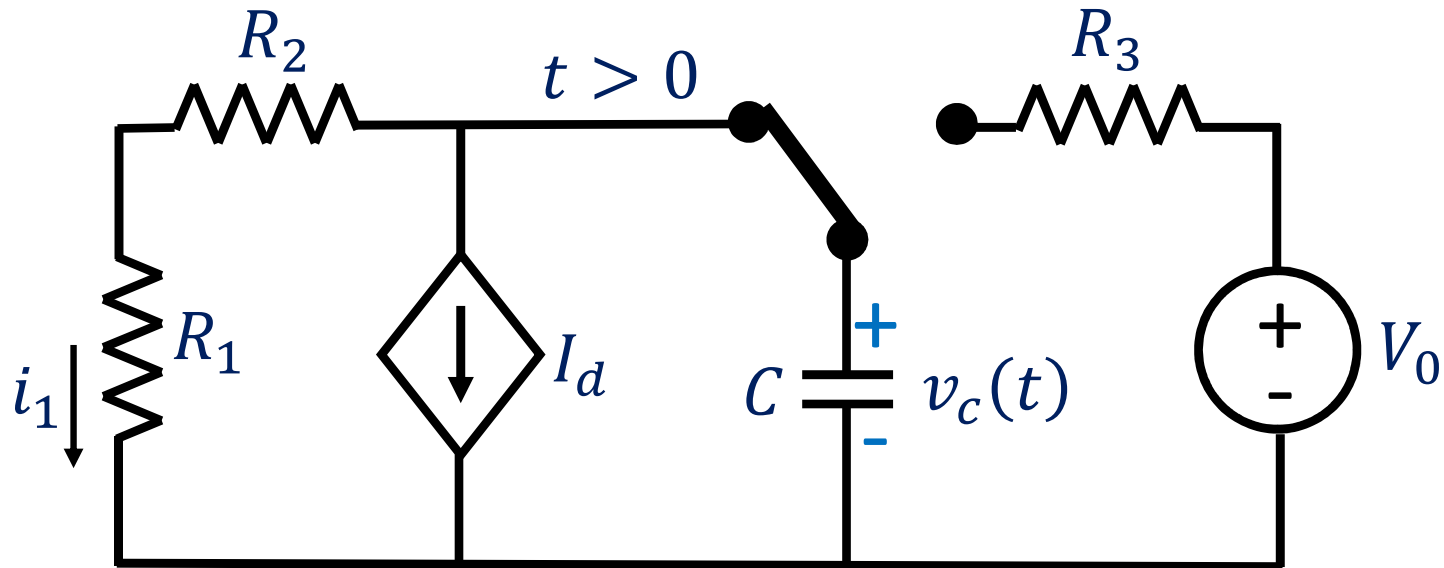
RC Circuit with Dependent Source



$$\frac{v}{R_1 + R_2} + I_d = 1 \quad \Rightarrow \quad \frac{v}{30} + 1.5 \left(\frac{v}{30} \right) = 1 \quad \Rightarrow \quad v = 12V$$

$$R_{TH} = 12\Omega$$

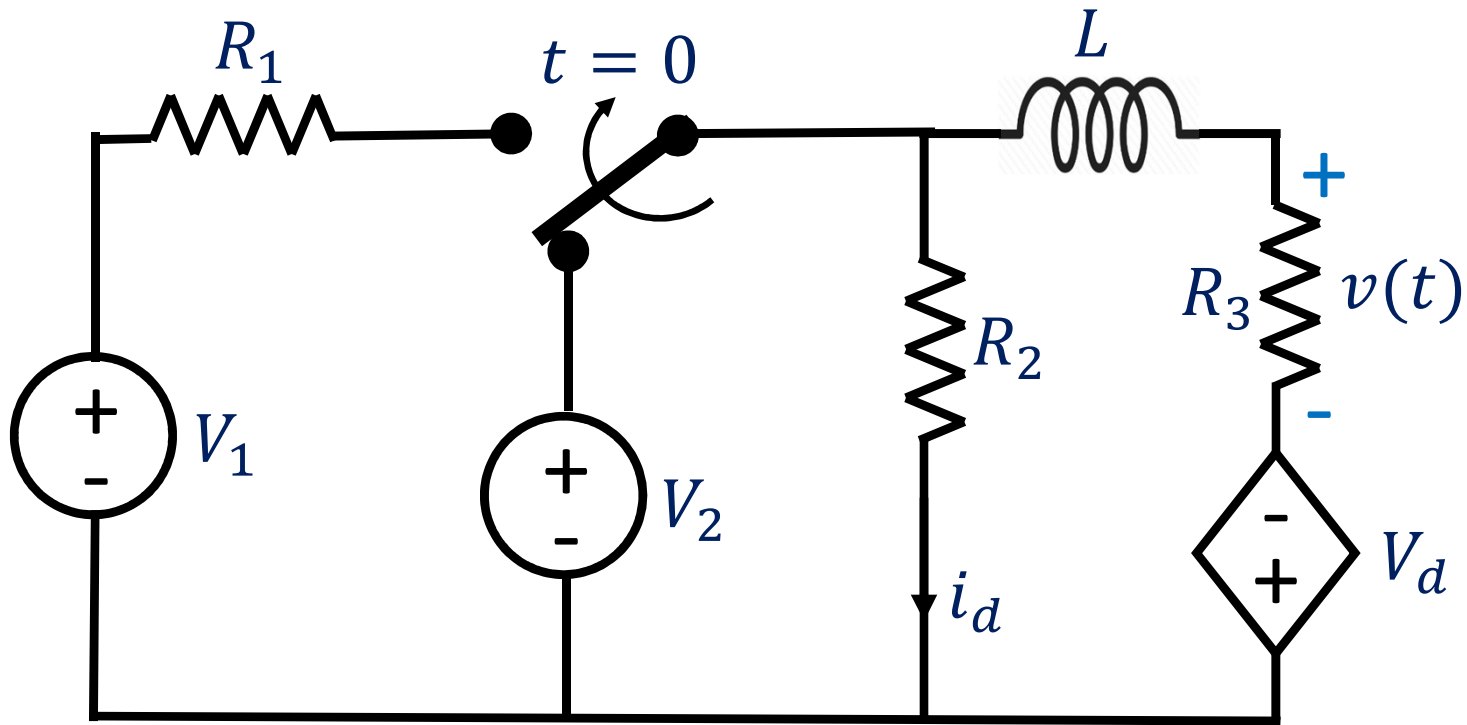
RC Circuit with Dependent Source



$$\tau = R_{TH}C = 12 \times 1 \times 10^{-6} \text{ s} = 12 \mu\text{s}$$

$$v_c(t) = v_c(0^-)e^{-\frac{t}{\tau}} = 2e^{-\frac{t}{12 \times 10^{-6}}} \text{ V}$$

RL Circuit with Dependent Source



$$v(t) = ?$$

$$R_1 = 2\Omega$$

$$R_2 = 4\Omega$$

$$R_3 = 6\Omega$$

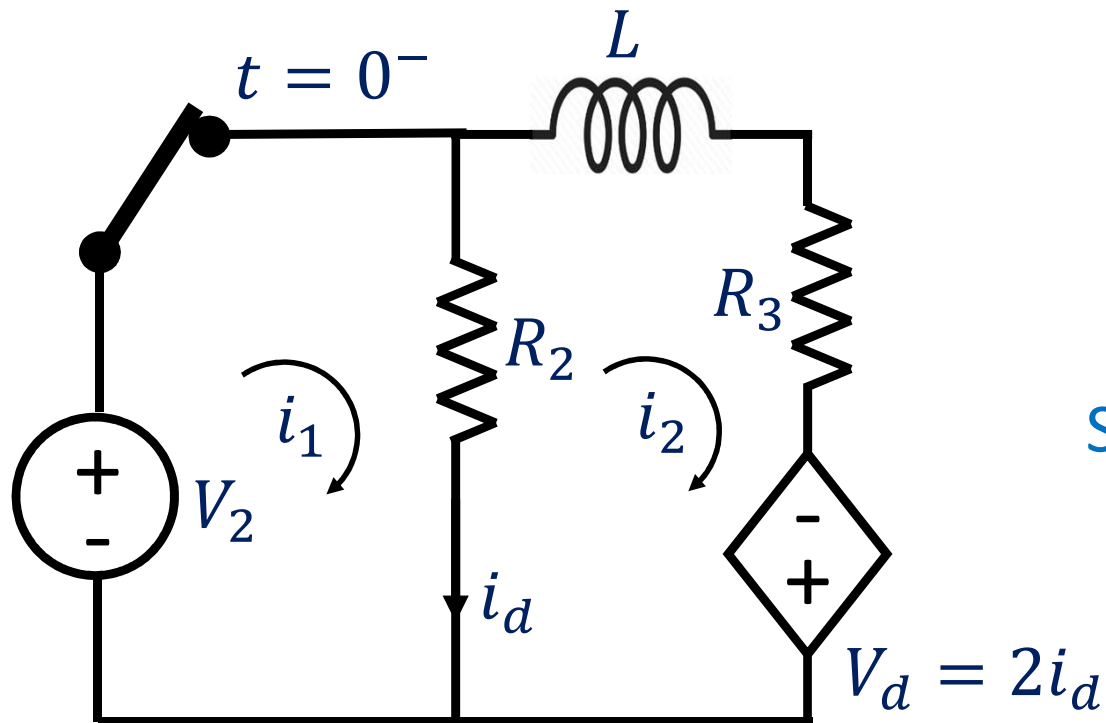
$$L = 3H$$

$$V_1 = 36V$$

$$V_2 = 12V$$

$$V_d = 2i_d$$

RL Circuit with Dependent Source

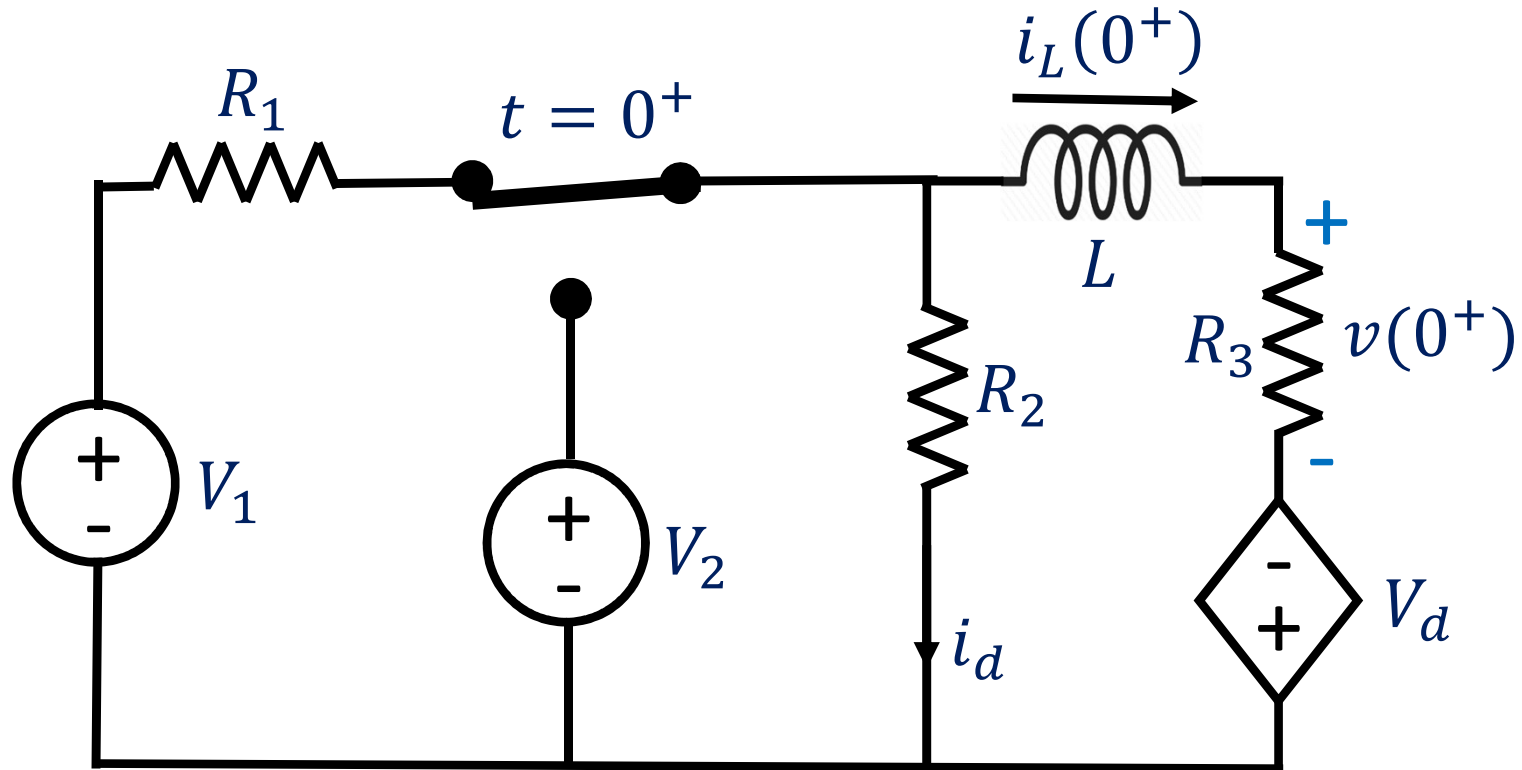


$$\begin{bmatrix} R_2 & -R_2 \\ -R_2 & R_2 + R_3 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_2 \\ V_d \end{bmatrix}$$

Solving Mesh Current Equations

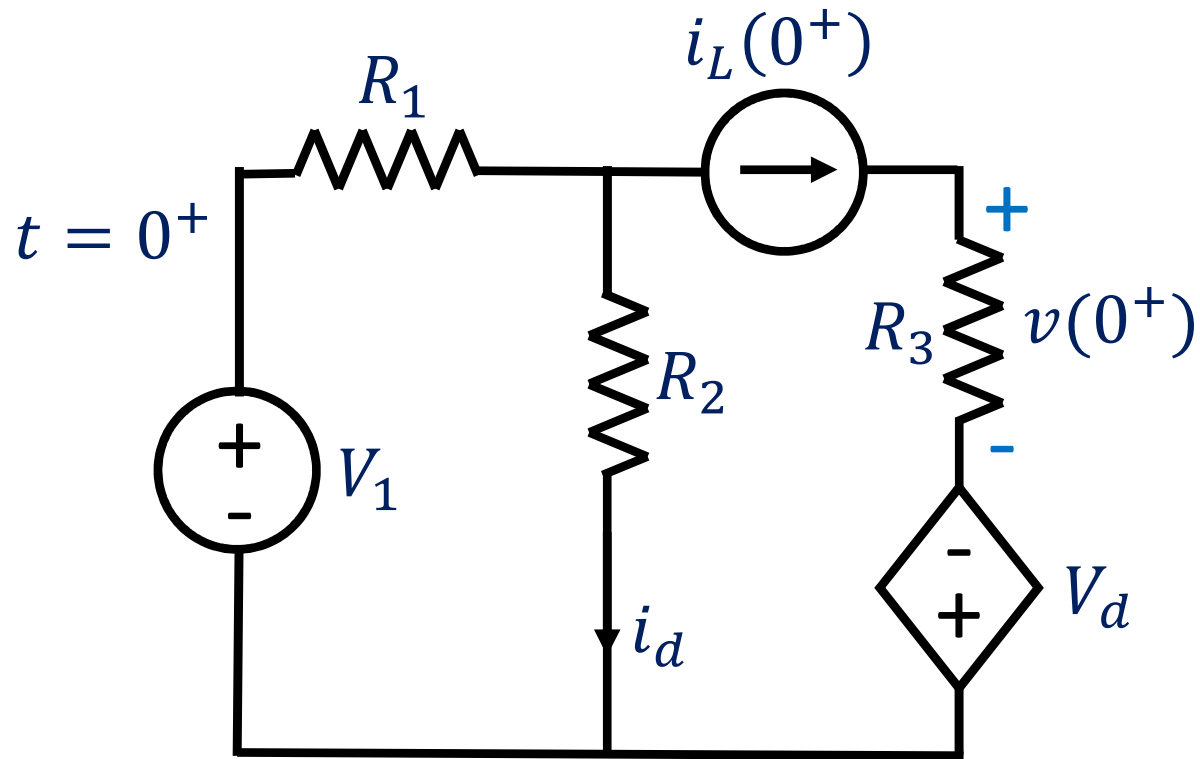
$$i_2 = i_L(0^-) = 3A$$

RL Circuit with Dependent Source



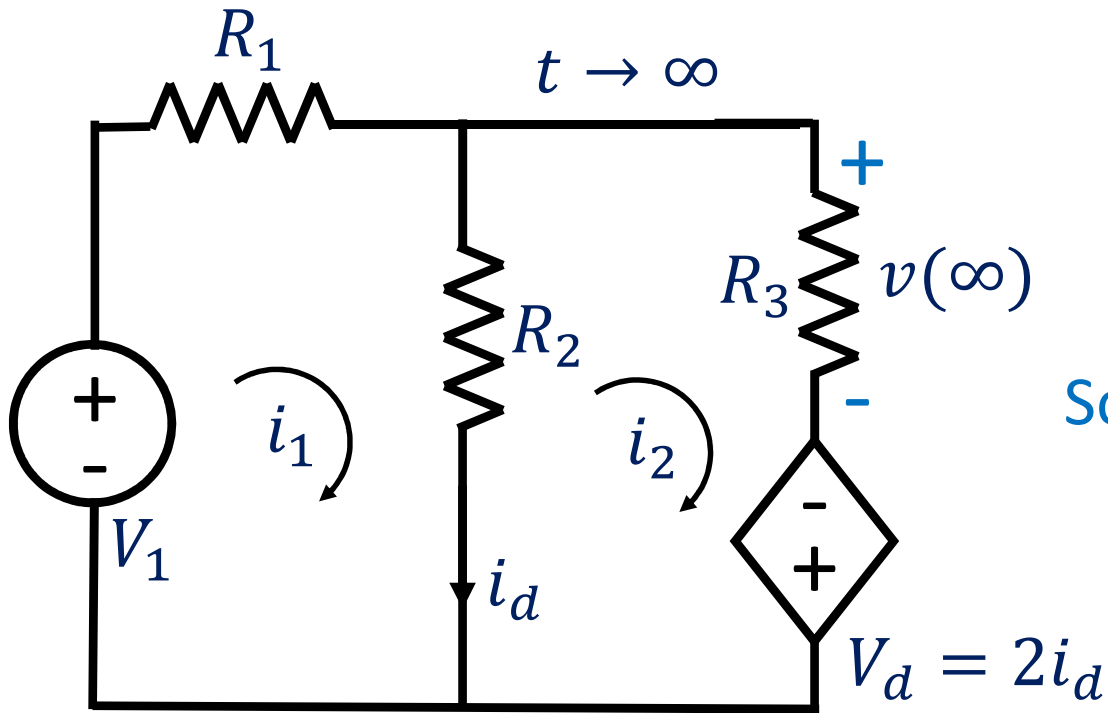
$$i_L(0^+) = i_L(0^-) = 3A$$

RL Circuit with Dependent Source



$$v(0^+) = R_3 i_L(0^+) = 6 \times 3V = 18V$$

RL Circuit with Dependent Source



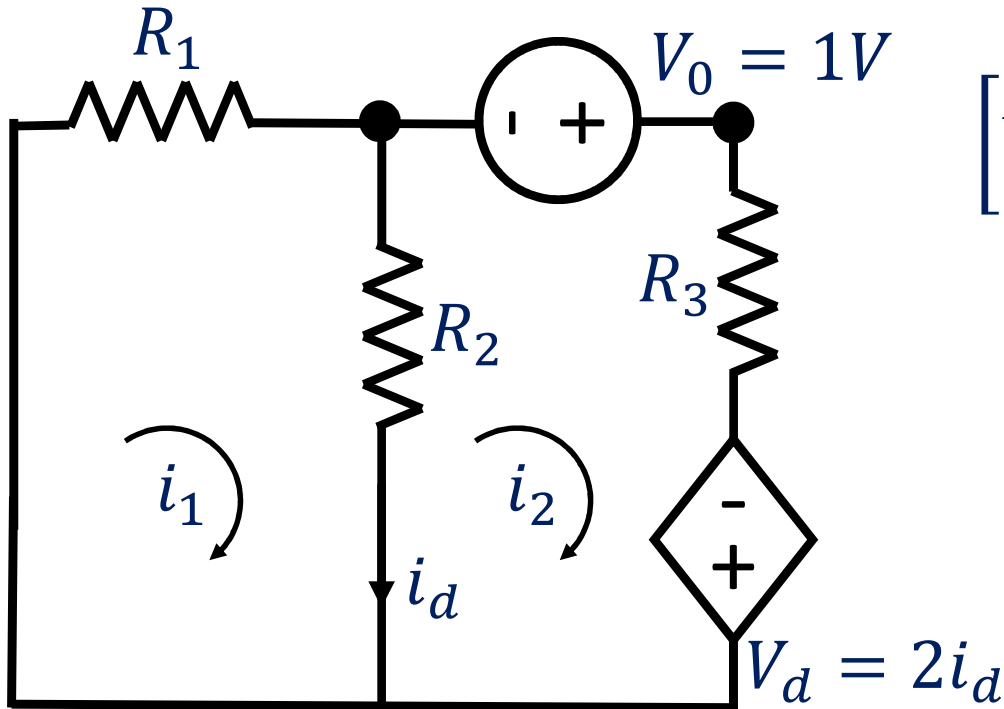
$$\begin{bmatrix} R_1 + R_2 & -R_2 \\ -R_2 & R_2 + R_3 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_d \end{bmatrix}$$

Solving Mesh Current Equations

$$i_2(\infty) = \frac{9}{2} A$$

$$v(\infty) = R_3 i_2(\infty) = 6 \times \frac{9}{2} V = 27V$$

RL Circuit with Dependent Source



$$\begin{bmatrix} R_1 + R_2 & -R_2 \\ -R_2 & R_2 + R_3 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 + V_d \end{bmatrix}$$

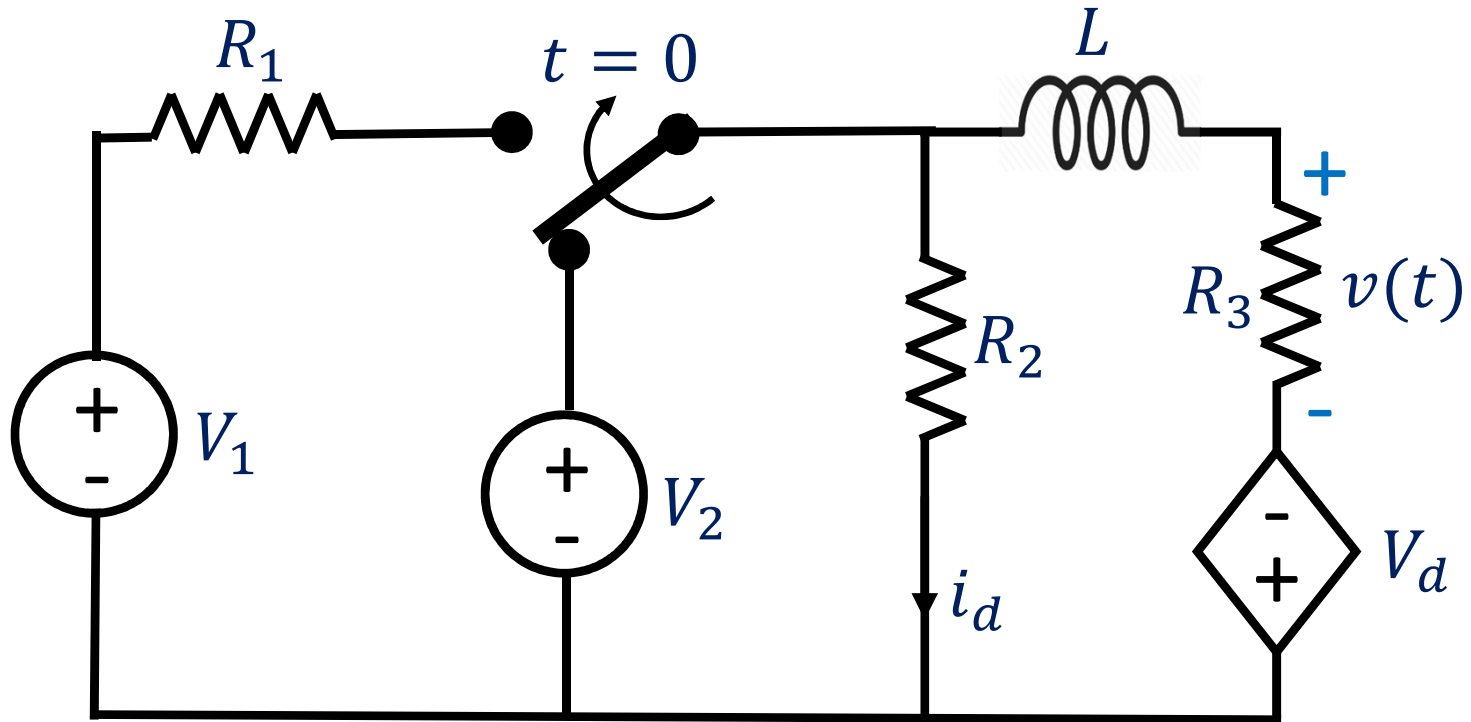
Solving Mesh Current Equations

$$i_2 = \frac{1}{8} A$$

$$R_{TH} = \frac{V_0}{i_2} = 8\Omega$$

$$\tau = \frac{L}{R_{TH}} = \frac{3}{8} s$$

RL Circuit with Dependent Source



$$v(t) = 27\left(1 - e^{-\frac{8t}{3}}\right) + 18e^{-\frac{8t}{3}}$$



Thank You